Resource-Aware Cattle Passage Counting with YOLO26n on a PYNQ-Z2 Vision Sensor

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***Abstract*—Cattle passage counting requires observations of the same animal at both ends of a channel. On an embedded device, a local detector call can recover a weak observation but delay the next global one. We deployed quantized YOLO26n on PYNQ-Z2 and evaluated that trade-off with wall-paced video. An initial configuration agreed with user-confirmed directional totals in two replays of each of four development clips (47 reference passages). In a subsequent 20-run ablation on the same clips, latest-frame selection with conditional local verification agreed with all four totals; the combined policy with history preference counted 11 against 12 on one clip. The history-enabled variants requested earlier images 15 times, but the age guard switched all requests to the latest image. The four latest-frame runs with local verification met the 1-s event-publication limit, with a worst age of 870.9 ms from scheduled frame arrival. These results identify conditional verification as the next comparator and leave the history-selection benefit untested. Directional totals do not establish event precision or recall; an independent timed reference and a controlled call budget are required.**

***Index Terms*—Cattle passage counting, visual sensing, edge computing, FPGA, PYNQ-Z2, YOLO26.**

I. Introduction

CATTLE passage records connect animal movement with farm operations. A fixed camera observing a controlled passage can provide a visual count without requiring every passage to yield a successful electronic identification read. The immediate sensing objective is to recognize a complete traversal and its direction. Assigning a temporary tracking identifier supports this objective; identifying a persistent individual animal is a separate recognition task.

Top-view cattle detection and tracking already support counting and identification research. CattleEyeView provides a multitask top-view dataset and tracking/counting baselines [1]. Other studies combine cattle detection, tracking and visual identification across environments or multiple cameras [2], [3]. Consequently, using a newer detector in this setting does not establish a research contribution by itself. The systems question considered here is how to preserve sufficient passage evidence when sensing and computation operate at different rates.

PYNQ-Z2 combines an embedded processor with programmable logic [4]. Zynq-based YOLO acceleration is also an established research direction [5]. In a complete counting system, however, additional work occurs around neural inference: decoding, crop/rotation, input preparation, short-term association, local verification and event publication. A verification call can improve a weak detection while delaying the next global observation. The resulting loss of passage evidence is not characterized by a detector-only FPS measurement.

We examine whether conditional local measurements help preserve the evidence needed to count a complete passage on a constrained device. The implementation uses YOLO26n, a ByteTrack-derived tracker and the existing O8 backend. It records the image selected for each observation and the calls spent on it. The development experiments test engineering feasibility and whether the proposed selection policy actually changes the observations.

II. Related Work

Top-view observations expose body appearance and reduce some side-view occlusions, but closely spaced animals and changing detection confidence remain relevant. Existing cattle studies address detection, temporary tracking and persistent visual identity with different labels and evaluation units [1], [2], [3]. We retain these distinctions when defining the output of the present system.

ByteTrack uses both high- and low-confidence detections for association [6]. A weak detection may therefore be useful for maintaining an existing track even when it is unsuitable for initiating a new track. Combining a global weak measurement and a strong local measurement introduces a further requirement: two views of the same animal must not become observations of two different tracks in one image. At the same time, a genuine weak neighboring animal must remain available for association.

YOLO26 supports the one-to-one export path used by this implementation [7]. The deployed compiler explicitly selects this path and checks a single-class output layout. This describes the present artifact, rather than every default prediction mode of the model family. Comparisons with other edge systems must also account for weight versions, input fields of view, quantization and timing boundaries.

III. System and Passage Definition

The data path comprises capture, source-coordinate geometry, global detection, optional local verification, short-term tracking and passage publication. The processing system handles capture, preprocessing and counting around the O8 convolution accelerator. The runtime records the graph, execution plan, wrapper and firmware identities so that a change in any of these assets can be distinguished from a change in the observation policy.

Let p_t denote the measured detection-box center, used as a torso-center proxy, along the calibrated passage axis. Two separated endpoint zones are defined by $a < b$: $A = \{p < a\}$ and $B = \{p > b\}$. A complete event requires a confirmed

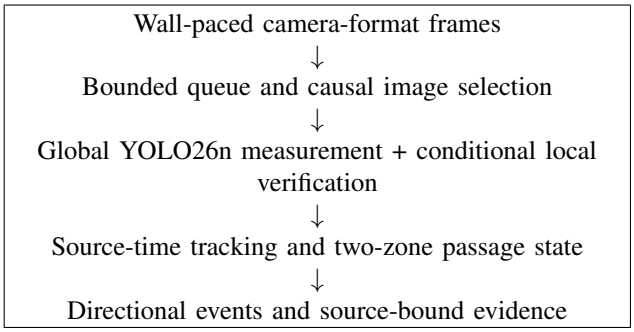


Fig. 1. Prototype sensing pipeline. Current development experiments replay archived frames at wall-clock speed. Physical camera transport is a separate qualification step.

short-term track with actual positions in both zones on different source images, in temporal order. A track first observed in the middle has no established origin. A passage ending before the second zone is reached remains censored or uncertain. Predicted positions support association and scheduling, but do not satisfy the event condition.

The development preset uses $a = 0.4$, $b = 0.6$ and a maximum real-observation gap of two source seconds. A spatial polygon further restricts admitted detections. Camera-specific geometry, orientation and directional labels are configuration inputs; expected animal totals are absent from the runtime configuration.

IV. Resource-Aware Observation and Verification

A. Causal Frame Selection

Capture publishes frames into a queue bounded to two images. Selection can use the latest available image or retain an already-arrived earlier image when measured motion and the unfinished passage phase indicate that an endpoint observation could otherwise be lost. This policy uses past measurements and current arrival state. It does not access future video images or interpolate a counting observation.

The configured acquisition rate and the actual global-model observation rate are reported separately. The latter depends on processing time and local verification work. The model-call ledger includes global, local and warm-up calls, allowing computationally fair comparisons rather than comparisons of nominal target rates alone.

An explicit history-age option can override the history preference when a newer image has already arrived. Before dequeueing, the consumer compares the oldest candidate's age, measured from its expected arrival, with the configured limit. If that candidate is over-age and a newer alternative is available, the latest image is selected. Otherwise the original selection remains. The original candidate age and override are logged. The option is disabled by default; the new original-format qualification preset declares 0.3 s before replay. This limits an older-frame preference, rather than guaranteeing a 0.3-s age for every selected image or a final event deadline. An already-old latest image and all processed late observations remain in the timing ledger.

B. Conditional Local Measurements

Local verification is triggered by bounded recent evidence for a weak origin or an unfinished destination. The selected current image is cropped and passed through the real detector. Candidate detections remain subject to class, geometry, area, overlap and neighboring-animal checks. Origin and endpoint verification share at most one additional model call per selected image.

The reliability review introduced a consolidation check for global/local views. A weak global box is superseded only when it overlaps the accepted local box and is geometrically owned by the target more strongly than by competing recent measured histories. Competitor geometry includes the tracker's actual tentative and confirmed observations, because the recovery log alone omits a newly confirmed neighbor's tentative observation. Both the original global detections and superseded rows are logged. The tracking and counting update occurs once per selected source image.

The currently implemented ownership check uses overlap of at least 0.5 and a target-versus-competitor overlap margin greater than 0.05. These values and their selection status are retained in the experimental record. Their robustness remains a subject for the budget-matched experiments.

The combined birth/endpoint-verification path requires the observation-based motion mode, which retains the real tentative history used in competitor geometry. Unsupported combined configurations are rejected before capture or model initialization; a Kalman-only tracking configuration does not provide the same qualification.

C. Event Scope and Failure States

An event records its video, session, run, entry/completion source times and source-frame indices. The evaluation adapter rejects conflicting scope and run identities. Legacy records can be normalized only through an explicit option that records the original file hashes and each added scope field; raw historical logs remain unchanged.

A failed live-camera read is reported as a capture fault, rather than a successful end of file. Without a verified sensor exposure timestamp, live timing starts at host frame-retrieval completion, and blocking retrieval duration is reported separately. Archived CFR replay uses verified relative source PTS and known wall-clock release times. These boundaries are kept distinct when interpreting latency.

Capture publication is timestamped after the frame enters the locked queue. Waiting for that lock is included in capture lateness and is also recorded separately. Statistical snapshots detach their samples while holding the lock and compute percentiles afterward, so numerical reporting does not retain the publication lock. Controlled software fixtures verify both the timestamp boundary and lock release.

V. Implementation and Provenance

The selected deployment uses a 256-pixel square detector input and prepared 1280×720 MJPEG acquisition at 5 Hz. The registered one-to-one output has shape $[1, 300, 6]$, containing box coordinates, confidence and class. The archived

S256 program and resident plan identify the actual quantized computation. Earlier 192/256/320 experiments use different trained checkpoints and therefore do not constitute a controlled same-weight resolution ablation.

Reproducibility requires more than a neural graph hash. The execution record binds the counting package, selected configuration, input, wrapper modules, program/plan metadata, native library and accepted firmware artifacts. Python/PYNQ and numerical-library versions are recorded with the board boot identity. Model files and full experiment payloads are retained in the private server archive, while source and compact evidence remain available for inspection.

VI. Experimental Protocol

A. Data Roles and References

The detector was fitted on 6,809 Cows2021 images, with 1,571 images used for development validation [8]. Their recorded capture dates and exact JPEG digests do not overlap. All 128 calibration images match training entries. We verified that the deployed checkpoint matches the final training checkpoint and that the qualified QDQ hash identifies the structurally normalized graph. All 10,402 source-role entries use a folder value consistent with 100-index bins of the original image number. A bin can span capture dates, so it does not identify a video session. These checks establish file and split provenance; they do not establish distinct herds or exclude near duplicates.

The four counting clips are monitor footage from MultiCamCows2024 [9]: camera 1 on August 14 and 15, 2023, and camera 2 on August 19 and 20, 2023. The known-direction totals are 12, 12, 11 and 12. The initial August 20 evaluation counted 9/12. That clip entered development when its result was used for diagnosis. All four clips in the ablation are development material.

An event reference records complete traversals, direction, entry and completion times, and censored or disputed passages. Directional totals can agree even if a miss cancels a duplicate. A separate source-image check of predictions was recorded; independent timed references are still unavailable.

B. Acquisition Preparation

Historical preparation converted the selected source intervals to 1280×720 MJPEG at 5 Hz while retaining the full field of view. Region cropping, rotation, detector preparation, inference, association and counting run online on the board. The replay measurement excludes the earlier format conversion and does not qualify native high-resolution H.264 decoding or physical camera transport.

The prepared streams contain 403, 649, 405 and 599 frames. Camera 2 August 19 uses the requested 14–95-s window. Camera 1 August 14 includes 52 source frames of existing preroll before the user-reviewed 8912–10873 interval; completed preroll events are excluded from its reference. Source hashes, conversion commands and source-frame mappings are retained with the experiment manifests.

TABLE I
Selected Revision V4: Development Replays (Two Repeats)

Clip	Counts (reference)	Max event lag (ms)	Global rate (Hz)
Aug14	12,12 (12)	608.1	2.81–2.82
Aug15	12,12 (12)	627.6	2.89–2.92
Aug19	11,11 (11)	915.5	2.90–2.91
Aug20	12,12 (12)	891.0	2.80–2.90

TABLE II
1280×720 Development Replays (Two Repeats)

Clip	Counts (reference)	Max event lag (ms)	Global rate (Hz)
Aug14	12,12 (12)	633.5	2.80–2.82
Aug15	12,12 (12)	752.1	2.90–2.91
Aug19	11,11 (11)	1059.1	2.84–2.90
Aug20	12,12 (12)	924.9	2.89–2.90

TABLE III
Historical Development Replay Before the 2026-11-09 Reliability Revision

Clip	Count/ reference	Wall time (s)	Max lag (ms)
Camera1 Aug14	12/12	80.94	633.21
Camera1 Aug15	12/12	130.17	744.33
Camera2 Aug19	11/11	81.42	1026.63
Camera2 Aug20	12/12	120.18	861.55

C. Metrics and Controls

For reference event set G and prediction set P, matching requires the same video, session and direction within a predeclared time tolerance. The assignment first maximizes one-to-one match cardinality and then minimizes absolute timing error. With match set M,

$$\mathrm{TP} = |M|, \quad \mathrm{FN} = |G| - |M|, \quad \mathrm{FP} = |P| - |M|. \tag{1}$$

Precision, recall and F1 follow these event counts; undefined empty denominators remain explicit. Temporal candidate ambiguity and independent physical-direction correspondence are reported separately.

The exploratory comparison kept the detector, input, channel geometry and counting rules fixed within each clip. It ran five policies once per clip in a rotated order: a 2.5-Hz target without local calls (U0), latest-frame selection (L0), history preference (C0), latest-frame selection with local verification (L1), and the combined policy (C1). L0 through C1 used a 5-Hz target. These are target rates; contention and processing time determine the realized calls and images. U0 also changes the tracker's nominal clock and is a system baseline. Replays of the same clip do not add independent sessions.

VII. Current Development Evidence

Table III records the previous source version's aggregate evidence. The actual global observation rate was approximately 2.86–2.95 Hz. The August 19 event exceeded the original

1000-ms publication limit; its frozen service result remains a failure. The historical engineering delivery interpretation is recorded separately and is not used to relabel that contract.

Table II reports eight further 1280×720 replays, two per clip, with the measurement-ownership and scope revisions. All known directional totals agree. The first August 15 replay nevertheless reached 1541.132 ms observation age and 538.753 ms capture lateness, exceeding the frozen 1500-ms and 500-ms limits. The second August 19 replay reached 1059.061 ms event-publication age against the original 1000-ms limit. These runs remain timing failures. The worst per-run observation p95 upper bound was 809.5 ms, and stream drain stayed below 0.673 s.

The capture-lateness implementation in those historical runs sampled time before taking the publication lock; it therefore did not include lock waiting. A controlled fixture exposed this gap by imposing a 250-ms wait: the old meter reported 100 ms for a frame enqueued after 350 ms. A second fixture exposed percentile calculation under the same lock. The corrected revision passed 139 cloud tests and the same 139 tests on the target ARM environment. These software controls establish implementation behavior, rather than cattle accuracy or an explanation for every timing outlier.

A separately frozen joint candidate reduced full-field acquisition to 640×360 while retaining the 256-pixel detector, weights, normalized channel and counting rules. Every derived frame retained the relative CFR timeline, and integer crop coordinates were rescaled with explicit rounding. Both camera 1 clips still yielded 12 in each repeat. Camera 2 August 19 yielded 10 and 10 against 11; August 20 yielded 7 and 9 against 12. This candidate is rejected on count agreement. Its acquisition and runtime changes do not establish a single-factor explanation for the losses.

The selected revision restores the exact 1280×720 inputs and pixel geometry, with the explicit history-age preference in the corrected runtime. Its two controlled selection fixtures and all prior checks passed 141 cloud tests and 141 ARM tests. A persistent SFTP connection provided passive status reads, avoiding fresh board Python or SSH probe processes during inference. Table I reports the entire predeclared block of eight runs: each known directional total agrees, with no reported complete event in the opposite direction. The actual global observation rate is 2.80–2.92 Hz. All runs meet the original limits: event-publication age at most 1000 ms, observation p95 at most 1000 ms and maximum at most 1500 ms, capture lateness at most 500 ms, drain at most 1 s and a queue bounded to two images without future-frame access. The observed maxima are 915.546 ms event-publication age, 979.776 ms observation age, 253.119 ms capture lateness and 0.731 s drain; the largest per-run observation p95 upper bound is 841.6 ms. Publication age starts at the completing source image's scheduled arrival, rather than an independently annotated physical crossing time.

All source, configuration, input and execution-asset fingerprints were verified before and after each run and rechecked with the raw ledgers on the private server. Source-image panels for all 47 first-repeat events were inspected by AI reviewers without an obvious visual counterexample. Three images per event cannot establish continuous identity or exclude missed passages. The raw runtime status remains unscored without formal event references, and its older 500-ms timing flag is retained as false. These records coexist with the externally checked original 1000-ms contract; neither is rewritten. The selected block supports engineering feasibility on the four development clips. Changes in selection, capture metering and monitoring do not constitute a matched single-factor ablation. Independent event scores, power measurements and FPGA resource reports remain pending their actual records.

TABLE IV
EXPLORATORY ABLATION ON FOUR DEVELOPMENT CLIPS

Policy	Clip totals	E_Σ	Calls (G/L/W)	Max age (ms)
U0	11, 12, 9, 9	6	1030/0/4	432.4
L0	12, 11, 9, 10	5	1256/0/4	627.9
C0	11, 12, 10, 10	4	1261/0/4	631.4
L1	12, 12, 11, 12	0	1178/52/4	870.9
C1	12, 12, 11, 11	1	1187/48/4	929.5

Clip order: Aug14, Aug15, Aug19, Aug20; reference: 12, 12, 11, 12. E_Σ is the sum of absolute directional total errors. Calls are global, local and warm-up. Max age is the largest event-publication age across the four replays.

A. Exploratory Board Ablation

All 20 runs completed and met the original replay clock limits. We verified their terminal records and recomputed counts, selected source times and call totals from the frame and event ledgers. Table IV shows all count outcomes. L1 agreed with the known-direction totals in all four clips. C1 agreed in three clips and counted 11 of 12 on August 20, where the previous two V4 replays had both counted 12. This is a count discrepancy under a repeated development replay; it cannot be assigned to a specific missed traversal without a timed reference.

The age guard switched every requested earlier frame to the latest frame: two switches in C0 and thirteen in C1, with no history reads. Across the clips, C1 used +2, +1, 0, +2 total calls relative to L1, in chronological clip order. The totals are close, but the actual global and local call allocations and frame times differed. The comparison does not isolate the effect of history selection. The board export's embedded bitstream and HWH matched the frozen deployment files. The measured peak process RSS ranged from 198.5 to 209.5 MiB across these runs; the export contained neither an implementation report nor a routed checkpoint. Board power was not measured.

VIII. Discussion and Next Experimental Decisions

Latest-frame selection with local verification is the most direct candidate for the next comparison. On these clips it used fewer global observations than L0 while making additional local calls, and its directional totals agreed more often. Local verification can therefore change the count and the allocation of observations; the present comparison cannot establish how many event errors it prevents. If the history policy is retained, a prospective test must include cases in which it actually selects older images. The main evaluation needs independently

annotated crossings, at least one source session excluded from development, and a budget defined before the comparison.

Wall-paced replay exercises decoding, processing and event publication on the board. Its clock starts at a prepared frame's scheduled arrival. Camera exposure, transport and native capture must be measured separately before reporting physical end-to-end latency.

IX. Conclusion

A quantized YOLO26n counting pipeline on PYNQ-Z2 processed four development videos at wall-clock speed and agreed with their directional totals in the initial qualification. In a 20-run comparison on the same clips, latest-frame selection with local verification agreed on four clips, while the combined policy disagreed on one. The earlier-frame preference selected no history frames under the chosen age guard. Independent event scoring and a controlled budget are needed to determine whether conditional verification improves passage counting beyond these clips.

Author-Completion Items

Author details, contributions, funding, animal-data/ethics applicability, dataset permissions and the final generative-AI disclosure require confirmation by the actual authors before submission. AI assistance has been used for literature organization, manuscript drafting and software review; the project retains these activity records. No editorial submission or publication payment has been made.

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